

A Cyber Security Perspective on Quantum Computing



Along with the immense promise of quantum computing come the enhanced security threats too. This has opened up a whole new field known as post quantum security.

In his famous book ‘Across the Frontiers’ Werner Heisenberg wrote, “Not only is the Universe stranger than we think, it is stranger than we can think.” This quote is applicable to quantum computing, because of its promise and ability to solve otherwise non-solvable, complex problems in the classical computing world.

It’s not that quantum computing promises to speed up the processing of every individual instruction, thus defying Moore’s Law. But it promises to quickly solve the exponentially repetitive tasks, e.g., finding out two prime factors

of a given big integer number. This makes it a potential threat to the known cyber security universe – a threat to a computer’s hardware, software and stored information. It’s a potential threat to our existing cyber security concepts and techniques.

In this article we are going to review such a threat to gain a deeper understanding about it; we can then arrive at our own opinions, independently.

Figures 1 and 2 indicate the potential of the so-called quantum threat and the global investments in quantum computing, with China leading the pack.

Existing security techniques

Let’s look at the current cyber security techniques. One obvious method of cracking an encryption code is brute force. A hacker can keep on trying all possible combinations till the code is broken. This method is not practical as it requires years to try out all combinations of the key. Currently, security across networks is based on a public key encryption system. Some well-known techniques are Diffie-Hellman, elliptic curve, RSA, etc.

Let us look at RSA. It is related to getting the private key by continuously factoring the public key, a big integer,